

Institute of Engineering & Technology, Bundelkhand University Jhansi

B. Tech (CS) Semester I

Computer Concept & Programming in C

Sessional-I 22/12/2023

Time: 30 Minutes

Maximum Marks: 15

Multiple Choice Questions.

- Q1. The 2's complement of binary number 0.0101 is (a) 1.10101 (b) 1.10100 (c) 0.10101 (d) 0.10100 1
- Q2. For storing a character we required (memory space) (a) 2 bytes (b) 4 bytes (c) 1 byte (d) 8 bytes 1
- Q3. The expression  $x = 5 + \frac{3}{6} - 6$  evaluates to? (a) 8 (b) -8 (c) 5 (d) 0 1
- Q4. Which of the following statement is wrong? (a) `des = 523.36;` (b) `con = 'I * B';` (c) `this = K * 50;` (d) `7 + x = sum;` 1
- Q5. While purchasing certain items, a discount of 10% is offered if the quantity purchased is more than 100. If quantity and price per item are input through the keyboard, Draw a flow chart to calculate the total expenses. 2
- Q6. If a four-digit number is input through the keyboard, write a program to obtain the average of the first and last digit of this number. 2
- Q7. Convert  $(4A5.75)_{10} = (\text{-----})_2$ . 2
- Q8. Explain in detail various function of operating System. 5

OR

Print the following pattern using loop control statement.

(1,1) 0  
(2,1) 1 0 (2,2) 1  
(3,1) 0 1 0 (3,2) 0 (3,3) 1  
(4,1) 1 0 1 0 (4,2) 0 1 0  
(5,1) 0 1 0 1 0  
5 1 5 2 5 3 5 4 5 5

0.01001  
1.10100  
+1  
1.10101

0.01011  
1.10100  
+1  
1.10101  
1000 3469  
346  
n/10 = 34



Part- A: Note: Attempt all questions. Each question carry 1 marks.

1. Convert number  $(43.12)_5 = ( )_7$   
a) 23.165      b) 32.561      c) 23.561      d) 32.165
2. minimum number of temporary variables needed to swap the content of two variable is  
a) 1      b) 2      c) 3      d) 0
3. if integers need two byte of storage then maximum value of signed integer is  
a)  $2^{16}-1$       b)  $2^{16}$       c)  $2^{15}-1$       d)  $2^{15}$
4. Which of the following is a not a keyword in C language?  
a) Void      b) Volatile      c) ☒ Sizeof      d) Getchar
5. How many different solutions will be possible for  $(123)_5 = (x8)_7$ .  
a) 1      b) 2      c) 3      d) 4

Part- B: Note: Attempt any two questions. Each question carry 2.5 marks.

1. Define operating system and its functions.
2. Define compiler and interpreter.
3. Define Unary and Ternary operators in C with example.

Part- C: Note: Attempt any ONE questions. Each question carry 5 marks.

1. Write a program in C to print the sum and average of any five numbers.

OR

2. Write a program in C to print the area of a circle.

**Bundelkhand University, Jhansi**  
**Institute of Engineering and Technology**

**B. TECH (CS) (2023-24)**  
**1<sup>st</sup> YEAR/1<sup>st</sup> SEMESTER**

**SESSIONAL -2<sup>nd</sup>**

**Engineering Mechanics**

**Max Time: 1 Hour**

**Max Marks: 15**

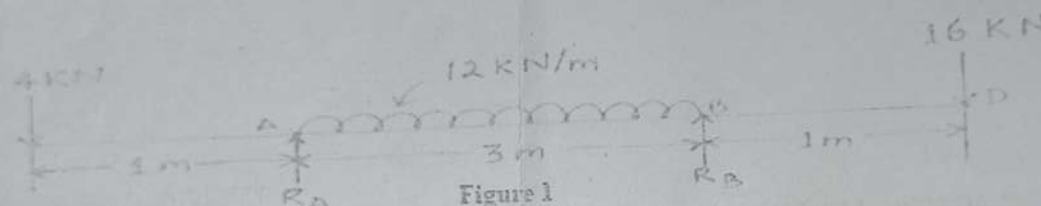
**Note-Attempt any three questions. Each carry five marks**

**3@5=15**

**Q1-** A uniform ladder of weight 250N act at CG and length of 5m is placed against a vertical wall in a position where its inclination to the vertical is  $30^\circ$ . A man weighing 800N climbs the ladder. At what position will he induce slipping? Take  $\mu = 0.2$  at both the contact surfaces of the ladder.

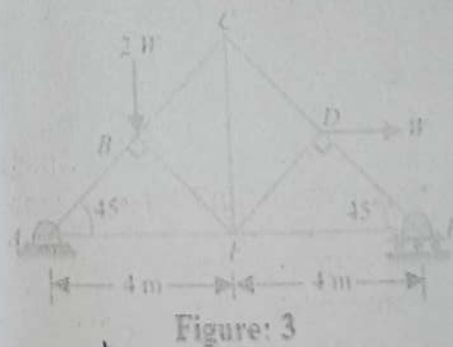
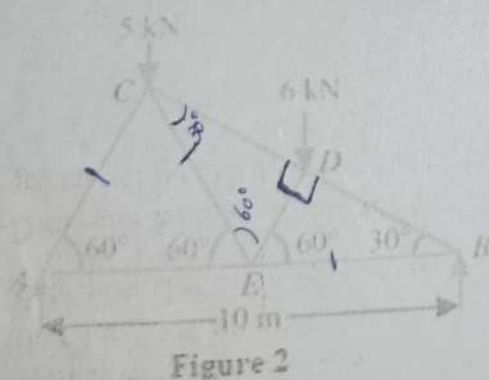
**Q2-** A man weighing 800 N dives into a swimming pool from a height of 20 m. It is estimated that he goes down in water by 2 m and then starts rising. Neglecting air resistance, determine the average resistance offered by water to the motion of the man.

**Q3-** Draw the SFD & BMD for the simply supported over-hanging beam shown in Figure 1 given below. Also, locate the point of contra flexure if any



**Q4-** A truss of Span 10 m is loaded as shown in Figure 2. Find the reactions and forces in the members of the truss.

**Q5-** A truss of Span 8 m is loaded as shown in Figure 3. Find the forces in the member's CD, FD, and EF of the truss.





Name & Roll no:  
B.Tech (computer science)

Engineering Mechanics (ME-101)

Max. Marks -15

Time- 60 min

Q1. Which of the following relation is true regarding magnitude of friction? (1)

1. Sliding friction > Rolling friction > Static friction
- ✓ 2. Static friction > Sliding friction > Rolling friction
3. Static friction > Sliding friction = Rolling friction
4. More than one of the above

Q2. Which of the following are contact force/forces? (1)

- I. Friction force
- II. Magnetic force
- III. Gravitation force
- ✓ 1. Only I
2. Both II and III
3. Only III
4. Only II

Q3. What is Kinematics? (1)

- 1) Branch of dynamics which deals with the study of bodies under motion without considering forces
  - 2) Branch of dynamics which deals with the study of bodies at rest
  - 3) Branch of dynamics which deals with the study of bodies under motion by considering forces
  - 4) Branch of dynamics which deals with the study of bodies under the motion
- Q4. Equilibrant is equal and opposite to the ... of several forces acting on a body. (1)

Q5. Define - (i) Law of Parallelogram (ii) Principle of Transmissibility (2)

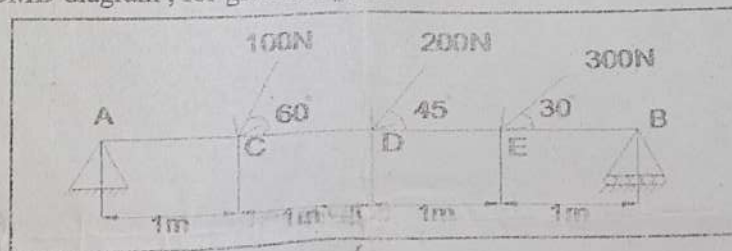
Q6. Determine the magnitude and direction of the resultant of the following set of forces acting on the body-

- (i) 200 N inclined  $30^\circ$  with east towards north
- (ii) 250 N towards the south
- (iii) 300 N towards North West, and
- (iv) 350 N inclined at  $40^\circ$  with west towards south

What will be the equilibrant of the given force system? (2)

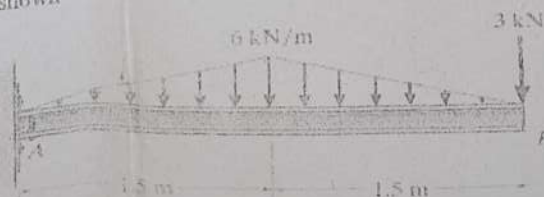
Q7. Define - (i) Angle of Friction (ii) Angle of Repose (2)

Q8. Draw SFD and BMD diagram, for given diagram. (2.5)



Q9. Derive Expression For Ratio of Tension On Tight Side and Slack Side. (2.5)

Q10. A Beam has been loaded and supported as shown in fig. Find the support reaction. (2.5)



ECE)  
(2nd Sem) (Electrical)

1. Define (i) Cycle; (ii) Frequency (iii) phase (iv) amplitude, (v) Time Period.
2. Define (i) Peak factor (ii) form factor (iii) ~~relative~~ reactance. (iv) Resonance (v) Impedance.
3. Write advantages of three phase system?



उपासना शर्मा

B.TECH(1<sup>st</sup> semester) EC-1862

1<sup>st</sup> sessional test

Electronics engineering

M.M : 15

Time : 1 hr

1. Differentiate between P and N type semiconductor. [2]
2. What is Filters? State different types of filters. [3]
3. Define ripple & ripple factor for HWR. [2]
4. Define TUF. [1]
5. Draw the circuit diagram of voltage multiplier circuit [3]
6. Draw  $v_i$  characteristic of pn junction diode. [2]
7. How the Zener diode act as voltage regulator [2]

B.TECH(1<sup>st</sup> semester) EC-1848

1<sup>st</sup> sessional test

Electronics engineering

M.M : 15

1. Differentiate between P and N type semiconductor.
2. What is Filters? State different types of filters .
3. Define ripple & ripple factor for HWR
4. Define TUF.
5. Draw the circuit diagram of voltage multiplier circuit
6. Draw v i characteristic of pn junction diode
7. How the Zener diode act as voltage regulator

Institute of Engineering & Technology, Bundelkhand University Jhansi

Name .....

Branch .....

Sessional 1<sup>st</sup> 2023-24

B.Tech (All Branches)

Engineering Physics

Semester-1

Time: 1 hour

Max.Marks:15

1. Derive time dependent Schrodinger wave equation.
2. Give a brief description of Davisson and Germer experiment.
3. Explain Heisenberg's uncertainty principle, and discuss any one application.
4. Prove that relation between Phase velocity and Group velocity.
5. Calculate the velocity and kinetic energy of a neutron having de-Broglie wavelength  $1.2 \text{ \AA}$  or

Describe and Prove that Compton Effect?



Institute of Engineering & Technology, Bundelkhand University Jhansi

Name .....

Branch .....

Sessional II 2023

B.Tech (All Branches)

Engineering Physics

Semester-1

Time: 1 hour

Max.Marks:15

1. Derive time dependent Schrodinger wave equation.
2. The Compton effect is associated with: ~~(a) visible light~~  
☒ (b) gamma rays (c) beta rays d) positive.
3. Explain Heisenberg's uncertainty principle, and discuss any one application.
4. Prove that relation between Phase velocity and Group velocity.
5. Calculate the velocity and kinetic energy of a neutron having de-Broglie wavelength  $1 \text{ \AA}$ .  
Or Describe and Prove that Compton Effect.

B

- 1) find the value of the  $\lambda$  &  $\mu$  so that the system  
 $2x + 3y + 5z = 9$ ,  $7x + 3y - 2z = 8$ ;  $2x + 3y + \lambda z = \mu$   
has (i) unique sol (ii) infinite (iii) no sol<sup>n</sup>.
- 2) find Eigen Value, Eigen vector, verify Cayley Hamilton  
th. for matrix A then find  $A^{-1}$  when  $A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -4 \\ -1 & -2 & 0 \end{bmatrix}$
- 3) find rank  $A = \begin{bmatrix} 1 & 2 & 3 & 2 \\ 2 & 3 & 5 & 1 \\ 1 & 3 & 4 & 5 \end{bmatrix}$



29. Expand  $e^x \cos y$  near the point  $(1, \pi/4)$  by Taylor's th.

30. Test for consistency & solve  $x + 5y + 7z = 15, 2x + 3y + 4z = 11,$

$$x - 2y - 3z = -4, 3x + 11y + 13z = 25$$

31. Find rank by echelon form  $A =$

$$\begin{bmatrix} 1 & 2 & 2 & 2 \\ 2 & 3 & 5 & 1 \\ 1 & 3 & 4 & 5 \end{bmatrix}$$

1082.  $y = \sin(2\sin^{-1}x)$  prove that  $(1-x^2)y_{n+2} = (2n+1)xy_{n+1} +$

$$\frac{9-5x^2}{4} y_n$$

$$30 \frac{11}{4}$$

$$29$$

$$\frac{15}{4} \frac{11}{4}$$

$$11 \frac{15}{4} \frac{11}{4} \frac{15}{4}$$

$$\frac{22}{22} \frac{20}{22} \frac{18}{22} \frac{16}{22} \frac{14}{22} \frac{12}{22} \frac{10}{22} \frac{8}{22} \frac{6}{22} \frac{4}{22} \frac{2}{22}$$



for n

$$\frac{d}{dx} \left( \frac{y(1+y^2)}{x^2} \right) + \frac{y(1+y^2)}{x^2} = \frac{y(1+y^2)}{x^2} (1+y^2)$$

QD

Saathi

Date: / /

Set A

1) If  $u = \tan^{-1} \left( \frac{x^3 + y^3}{x - y} \right)$  prove that  $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \sin 2x$

2)  $y = (\tan^{-1} x)^2$  prove that  $(1+x^2)y'' + 2x(1+x^2)y' = 2$

3) If  $y = \sin(m \sin^{-1} x)$  prove that  $(1-x^2)y'' - 2xy'y'' + m^2 y = 0$

4) Expand  $x^y$  in power of  $(x-1)$  &  $(y-1)$  by Taylor's th.

$$\delta(x,y) = \delta(x,y) + \frac{\partial}{\partial x} \left( \frac{x^2 y}{2!} \right) + \frac{\partial}{\partial y} \left( \frac{x^2 y}{2!} \right) + \frac{1}{2!} [x^2 y + y^2 x + x^2 y]$$



cesuanchlonide

Basic Electrical Engineering  
2nd sessional Exam  
EC/BT 1 sem.

Max. marks 15

Attempt all questions. Each question carries equal marks.

Q1. define: Magnetic flux, MMF, leakage flux, fringing, reluctance.

Q2: Explain RLC series circuit.

Q3. define bandwidth and selectivity.

Q4. Write advantages of three phase system over single phase system.

Q5. define : active power, reactive power, apparent power.

Institute of Engineering & Technology, Bandeikhana University, Hanoi

B. Tech (CS) Semester I

Computer Concept & Programming in C

Sessional-II 11/01/2024

Time: 60 Minutes

Maximum Marks: 15

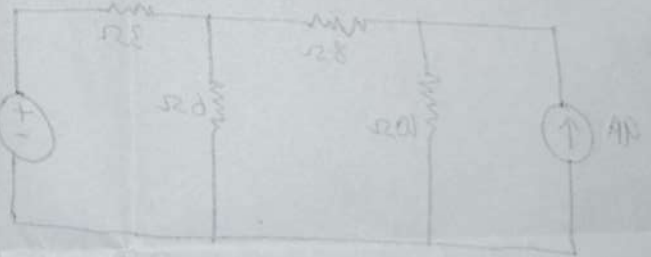
Note: Solve any three.

- Q1. What is a data type? Explain the four basic data types of C.
- Q2. Write a C program to find the real roots of the quadratic equation  $ax^2 + bx + c = 0$ .
- Q3. How do you classify the control statements? Explain with suitable example.
- Q4. In the process of grouping the people into CHILD, YOUNG and OLD categories, depending on the age group, which control statement would you like to choose and why? Write C program for it.
- Q5. Explain storage classes in detail.
- Q6. Write a C program for binary search.



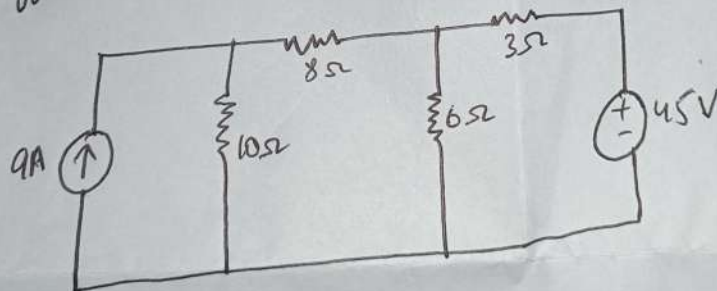
Chemistry (1<sup>st</sup> Semester) (1<sup>st</sup> Sessional)

1. Explain MOT with  $N_2$ .
2. Bragg's Equation.
3. Differentiate between Smectic and Nematic Liquid Crystals.
4. Formation of following compounds by Grignard reagent:
  - i) Sodium Acetate.
  - ii) Benzoic Acid.
  - iii) Acetophenone.



Electrical (ECE) (1<sup>st</sup> Semester) (1<sup>st</sup> Sessional)

1. Define 1) KCL 2) KVL 3) Ohm's Law 4) MMF  
5) Permeability.
2. Derive maximum power transfer theorem.
3. Explain Eddy Currents and B-H Curve.
4. Differentiate between Active & Passive Elements.
- 5.



Find current across  $8\Omega$  using superposition theorem.



ECE (II<sup>nd</sup> semester) 1<sup>st</sup> sessional  
Professional Communication

classmate

Date \_\_\_\_\_

Page \_\_\_\_\_

- Q1. Write synonyms of ① Arrogant ② Barrier.
- Q2. Write Antonyms of ① Abhor ② Adequate.
- Q3. Write Homophones of  
(A) Adapt, Adept, Adopt.  
(B) Altar, Alter.
- Q4. What is Horizontal communication?
- Q5. Write characteristics of Non-Verbal communications?
- Q6. What is the difference between Intrapersonal communication and Interpersonal communication?
- Q7. What is downward communication?



ECE (I<sup>nd</sup> Semester) (1<sup>st</sup> Sessional)  
Ecology.

Date \_\_\_\_\_  
Page \_\_\_\_\_

- Q1. Energy flow in an ecosystem is always  
a) Unidirectional b) reversible c) cyclic d) None of these.
- Q2. The desert among the following which is not a coldest desert is  
a) Golbi Desert b) Atacama Deserts  
c) Patagonian Deserts d) Musaba Deserts.
- Q3. The darker zone in lake where light penetrates negligible is called the  
a) Euphotic Zone b) Lithoral Zone  
c) Limnetic Zone d) Benthic Zone
- Q4. Increase in fauna and decrease in flora would be harmful due to increase in  
a) Disease b)  $O_2$  c)  $CO$  d) radioactive
- Q5. The largest reservoir of Nitrogen in our planet is  
a) Ocean b) Biosphere c) Atmosphere d) fossil fuels.
- Q6. Food chain starts with  
a) Producer b) Carnivores c) Herbivores d) None of these

Write short note on:

7. Types of ecosystem (only flow chart with example).
8. Carbon cycle.
9. Ozone layer.





Dt. 05.12.2024

B.Tech. (E&C Engg.) Sem I  
Electronics Engineering Sessional 2  
Time: 12 - 01 PM

Max. Marks : 15

Note : Attempt all questions. Each carries 1 mark.

1. The gain of an ideal op amp is  $\dots 1 \dots$
2. The unit of slew rate is  $\dots \text{V}/\mu\text{s} \dots$
3. Bandwidth of an ideal op amp is  $\dots \infty \dots$
4.  $X + X = \dots X \dots$

Note : Attempt any 3 questions. Each carries 2 marks.

5. What are universal gates?
6. Define CMRR.
7. Define inverting amplifier.
8. Simplify  $Y = AB + (A + B)(\bar{A} + \bar{B})$ .

Note : Attempt any 1 question. Each carries 4 marks.

9. Explain op amp as integrator.
10. Simplify  $F = AB + AC + AD$ .

$$AB(C+\bar{C}) + A(B+\bar{B})C + A(B+\bar{B})(C+\bar{C})$$

$$AB(CD + C\bar{D} + \bar{C}D + \bar{C}\bar{D}) + A(BD + B\bar{D} + \bar{B}D + \bar{B}\bar{D})$$

$$ABCD + ABC\bar{D} + AB\bar{C}D + AB\bar{C}\bar{D} +$$

$$AB + (A+B)(\bar{A} + \bar{B})$$

$$AB + A\bar{A} + A\bar{B} + \bar{A}B + \bar{A}\bar{B}$$

$$AB + A\bar{B} + \bar{A}B + \bar{A}\bar{B}$$

$$AB + \bar{A}\bar{B}$$

Name & Roll no :  
B.Tech (computer science)

Institute of Engineering & Technology, Bundelkhand University Jhansi

Sessional 1<sup>st</sup>, 2022-23  
semester - 1<sup>st</sup>

Time - 60 min

Engineering Mechanics (ME-101)

M.M -15

- Q1. The unit of time is second which is a convenient fraction of the period by the earth's rotation i.e.  $1 \text{ sec} = 1/86400$  ..... (1)
- Q2. A Body which does not deform under the action of applied force is called the ..... (1)
- Q3. Fundamental Principles of Mechanics are ..... (1)
- Q4. Equilibrium ..... of several forces acting on a body ..... (1)
- Q5. Define - (i) Law of Conservation of Energy ..... (1)
- Q6. Define triangle law of vector addition ..... (2)
- Q7. State & Prove Lami's theorem ..... (2)
- Q8. Determine the magnitude and direction of the resultant of the following set of forces acting on the body - (2)
- (i) 200 N inclined  $30^\circ$  with east towards north
- (ii) 250 N towards the south
- (iii) 300 N towards North West, and
- (iv) 350 N inclined at  $40^\circ$  with west towards south.
- What will be the equilibrant of the given force system?

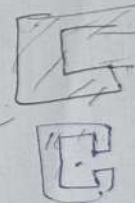


CT-2(BTech-EC ,1<sup>st</sup> Sem), Roll No/Name.....  
EC (11/01/ 2024)  
Max Marks: 10

Manufacturing Process  
Max Tim: 60 Minutes

NOTE- ATTEMPT ANY FIVE QUESTIONS

- |                                                       |        |
|-------------------------------------------------------|--------|
| Q1-DEFINE ELASTICITY AND PLASTICITY? ✓                | 2MARKS |
| Q2-GIVE PROPERTIES & APPLICATIONS OF GREY CAST IRON ✓ | 2MARKS |
| Q3- CLASSIFY MATERIALS USED IN MANUFACTURNIG ✓        | 2MARKS |
| Q4- WHAT IS MANUFACTURING PROCESS? ✓                  | 2MARKS |
| Q5-EXPLAIN PATTERN IN CASTING ? ✗                     | 2MARKS |
| Q6- WHAT DO YOU UNDERSTAND BY CASTING PROCESS? ✓      | 2MARKS |
| Q7-WITH NEAT SKETCH, EXPLAIN EXTRUSION PROCESS? ✓     | 2MARKS |
| Q8- EXPLAIN TOUGHNESS AND RESILIANCE ? ✓              | 2MARKS |



Institute of Engineering & Technology, Bundelkhand University, Jhansi

Name ..... Branch .....

Sessional 1<sup>st</sup> 2022-23

B.Tech (All Branches)

Engineering Physics

Semester-1

Time: 45 Min

Max.Marks:15

- Q.1 Obtain Einstein's mass energy relation, considering the variation of mass with velocity.
- Q.2 Discuss the mass variation according to special theory of relativity.
- Q.3 At what speed, the mass of an object will be double of its value at rest?
- Q.4 Calculate the rest mass, relativistic mass and momentum of a photon of energy 5eV.
- Q.5 Derive the relativistic formula for the addition of velocities.

Or

State the fundamental postulates of the special theory of relativity and hence deduce the Lorentz transformation



Department of Computer Science & Engineering  
Institute of Engineering Technology, Bundelkhand University

Semester : B.Tech(EC, BT, 2<sup>nd</sup> sem ) Year: First  
Sessionals : First

Subject : CCP  
Date: 08/03/2025

Attempt ALL MCQ(1\*5=5)

- Q1 What is computer memory?  
☒ A. device used to store information in computer B device to print output to screen  
C. device to process data D None of these
- Q2 Which of these are not memory devices of a computer?  
☒ A. Screen B. Cache C. Floppy D. All of these
- Q3 Which of these memory devices can be removed from the computer?  
A. Floppy B. RAM ☒ C. Hard disk D. Register
- Q4 What is RAM in a computer?  
E. Primary memory that stores data at random memory location  
F. Primary memory that stores data at contiguous memory location  
G. Detachable memory to upgrade computer's storage  
H. All of these
- Q5 Which of these is a category of RAM?  
A. Erasable RAM B. Dynamic RAM ☒ C. Programmable RAM D. None of these
- Attempt Any two
- Q6. Write a program to calculate area of triangle ?
- Q7 Write a program to calculate input number is even or odd?
- Q8. Write a program to calculate area of square.
- Q9. Write a program to calculate area of rectangle.
- over*

Institute of Engineering & Technology, B. U. Jhansi

Name .....

Branch .....

Sesional I<sup>st</sup> 2024-25

B. Tech (All Branches)

Engineering Physics

Semester - I

M.M. 15

Time - 2 hour

Q.1 Derive time dependent Schrodinger wave Equation.

Q.2 Prove that relation between Phase velocity and Group velocity.

Q.3 Explain Heisenberg's uncertainty principle and discuss any one application.

Q.4 X-rays have energy

(a) higher than Xrays (b) higher than ultraviolet rays  
(c) lower than visible rays (c) lower than infrared rays.

Q.5- Which of the following is not a particle characteristic  
frequency (b) mass (c) volume (d) kinetic energy



**INSTITUTE OF ENGINEERING & TECHNOLOGY, BUNDELKHAND UNIVERSITY, JHANSI**  
**B Tech (Electronics & Communication Engineering)**

ROLL No.....  
Subject: Manufacturing Processes (ME-1850)  
Time: 1 Hour

1<sup>st</sup> Year, I<sup>st</sup> SEM  
Sessional: II (2024-25)  
Max. Marks: 10

**Note:** Attempt any *five* Questions. Each question carries equal marks.

1. Explain the following terms with their sketches: (i) Ductility, (ii) Brittleness, (iii) Fatigue and (iv) Creep
2. Explain different types of cast iron.
3. Based on the percent of carbon, classify carbon steels. How do stainless steel become stainless?
4. What do you mean by case hardening? Explain different methods of case hardening.
5. Explain brass and bronze with their properties and applications.
6. What do you mean by forming operations? Explain in brief with examples and neat sketches.

INSTITUTE OF ENGINEERING & TECHNOLOGY, BUNDELKHAND UNIVERSITY, JH

B Tech (Electronics & Communication Engineering)

ROLL No. 30 .....

Subject: Manufacturing Processes (ME-1850)

Time: 1 Hour

1<sup>st</sup> Year,

Session

Max. M

**Note:** Attempt any *five* Questions. Each question carries equal marks.

1. What do you mean by Mechanical properties of materials? Explain any two properties in brief.
2. Sketch a stress-strain diagram for mild steel and explain each point concerning the sketch.
3. Write short notes on (a) Creep and (b) Fatigue.
4. Give a brief classification of materials.
5. What is wrought iron? Explain its properties and applications.
6. Explain cast iron and its properties. Also, enumerate different types of cast iron.



$$(D^2 - 2D + 5)y = e^{2x} \sin x$$

Pr. I.T.P. Q. 10

$$(D^2 + 1)y = \cot x \quad y = I.F. \int \frac{1}{D^2 + 1} dx$$

$$(D^2 - a^2)y = e^{-ax}$$

$$\frac{dy}{dx} + \frac{3}{x}y = \frac{1}{x^4}$$

Is given

$$\frac{dy}{dx} = P \quad \frac{dy}{dx} \quad \frac{dP}{dx} \quad \frac{1}{x^{n+1}}$$

$$\frac{dP}{dx} + \frac{3}{x}P = \frac{1}{x^4}$$

$$m(m-1) + m \cdot \frac{1}{x} + \dots = \dots$$

$$P + Q = \dots$$

$$m(m-1) + \dots$$

$$m = \dots$$

Institute of Engineering & Technology, Bundelkhand University Jhansi

Name ..... Branch ..... Sessional 2nd 2023-24 B.Tech (All Branches)

B.Tech (All Branches) Engineering Physics Semester-1 Max.Marks:15

- ✓ Q.1 Explain the colours when a thin film illuminated by white light is observed in reflected light.
- ✓ Q.2 Obtain the intensity expression for a single slit Fraunhofer diffraction pattern.
- ✓ Q.3 Define the polarization. Explain the double refracting crystal.
- ✓ Q.4 Explain and prove fiber optics and its type?
- Q.5 Prove the Diameter of Newton rings. Or Explain and describe construct reconstruction holography.

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Name & Roll no :

Sessional 2<sup>nd</sup>, 2022-23

Manufacturing Process

Time – 60 min

MM -10

All Questions carries two marks. (Do any five questions )

- Q1. Explain any one method of making metal powder.
- Q2. What is Difference between shaper and planer machine ?
- Q3. What is difference between hot working and cold working ?
- Q4. What is Brass? Discuss its applications .
- Q5. Discuss Oxyacetylene Gas Welding in details.
- Q6. Why are the basic requirement for choosing a plant location ? Explain different type of plant layout.
- Q7. Explain the following :  
Creep, fatigue, brittleness, hardness, toughness, rigidity, stiffness, elasticity

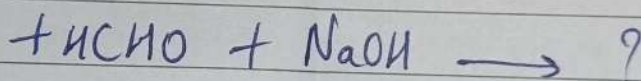
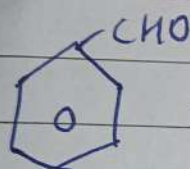
(ECE) (1<sup>st</sup> semester)  
(Chemistry) (2<sup>nd</sup> sessional)

classmate

Date \_\_\_\_\_

Page \_\_\_\_\_

1.



Write reaction mechanism.

2. Write MOT of NO.
3. Write reaction mechanism of dehydration of alcohol.
4. Write  $S_N1$  &  $S_N2$  mechanism reaction.



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Name .....

Branch .....

Sessional 1<sup>st</sup> 2023-24

B.Tech (All Branches)

Engineering Physics

Semester-1

Time: 1 hour

Max.Marks:15

Q.1 Obtain Einstein's mass energy relation, considering the variation of mass with velocity.

Q.2 Discuss the mass variation according to special theory of relativity.

Q.3 AT what speed, the mass of an object will be double of its value at rest?

Q.4 Calculate the rest mass, relativistic mass and momentum of a photon of energy 5eV.

Q.5 Derive the relativistic formula for the addition of velocities.

Or

State the fundamental postulates of the special theory of relativity and hence deduce the Lorentz transformation

$$\begin{array}{r} 3 \times 5 \\ 1.6 \\ \hline 0.8980 \end{array}$$
$$\begin{array}{r} 8 \\ 91 \end{array}$$
$$\begin{array}{r} -19 \\ -16 \\ \hline 25 \end{array}$$

$$\frac{wt + \delta + wt}{2} \quad 2 \cos \frac{wt + \delta - wt}{2}$$

$$\frac{2wt + \delta}{2} \quad 2 \cos \frac{\delta}{2}$$

$$2 \cos \frac{wt + \delta}{2}$$

$$2 \sin \left( \frac{wt + \delta + wt}{2} \right) \cos \left( \frac{wt + \delta - wt}{2} \right)$$

$$a \left( 2 \sin \frac{2wt + \delta}{2} \cos \left( \frac{\delta}{2} \right) \right)$$

Institute of Engineering & Technology, Bundelkhand University Jhansi

Name Muskan.....

Branch EEE....

Sessional 2<sup>nd</sup> 2024-25

B.Tech (All Branches)

Engineering Physics

Semester-I

Time: 45 Min

Max.Marks:15

Q.1 Explain and Derive Fresnel theory of optical rotation?

Q.2 Explain Fresnel and Fraunhofer diffraction?

Q.3 Explain interference of light and derive interference of reflected light?

Q.4 Explain Newton rings and derive diameter.

Q.5 Derive diffraction single slit. Or Derive Einstein coefficient.



$\frac{1}{2}$

$2 \sin \frac{\delta}{2} \cos \frac{wt}{2}$   
 $2 \cos \frac{\delta}{2} \sin \frac{wt}{2}$